Fimiston Open Pit (Super Pit)

The **Fimiston Open Pit**, commonly known as the **Super Pit**, is one of the **largest open-pit gold mines in the world**, located in Kalgoorlie, Western Australia. It is currently operated by Northern Star Resources. The mine is part of the historic Kalgoorlie goldfields and has been a major contributor to Australia's gold production for over a century.

🌍 Key Features of the Mine

1. 🏗️ Massive Open-Pit Structure

The Super Pit is approximately **3.5 km long, 1.5 km wide, and over 600 meters deep**.
👉 It is a classic example of **large-scale open-pit gold mining**.

2. 💎 Rich Gold Deposits

The mine extracts gold from the famous **Golden Mile ore body**, known for its rich mineralization.
👉 Important for studying **gold deposit geology and mineralization**.

3. ⛏️ Bench Mining Method

Mining is carried out using the **benching method**, allowing safe and efficient extraction of ore.
👉 Demonstrates **standard open-pit mining techniques**.

4. 🚜 Large-Scale Equipment

The operation uses massive **haul trucks, excavators, and drilling equipment** to handle large volumes of material.
👉 Highlights **high mechanization in mining operations**.

5. 🧪 Gold Processing Techniques

Ore is processed using **crushing, grinding, and cyanide leaching** to extract gold.
👉 Important for understanding **gold extraction and metallurgical processes**.

6. 🔄 Continuous Expansion and Optimization

The mine has undergone continuous expansion and optimization to increase efficiency and extend mine life.
👉 Demonstrates **long-term mine planning and resource optimization**.

7. 🌱 Environmental and Safety Practices

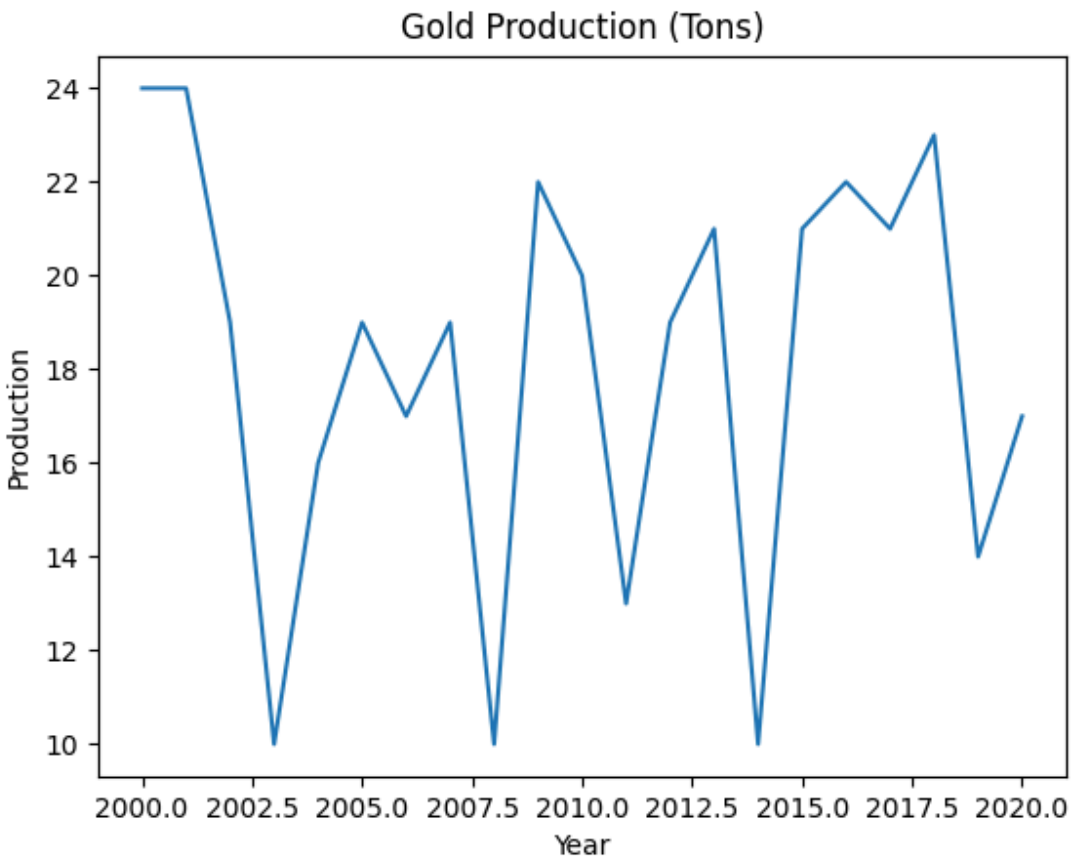
Modern operations include **waste management, land rehabilitation, and strict safety protocols**.
👉 Relevant for **sustainable mining and safety engineering**.

8. 📈 Economic Importance

The Super Pit is a major contributor to Australia's gold industry and global gold supply.
👉 It plays a key role in **mineral economics and international trade**.

📊 Chart-wise Description & Significance

1. Gold Production (Tons)



This chart shows the annual gold output of the Super Pit. Variations reflect operational efficiency, ore availability, and global gold demand.
👉 **Significance:** Helps in **production forecasting, reserve estimation, and long-term mine planning**.

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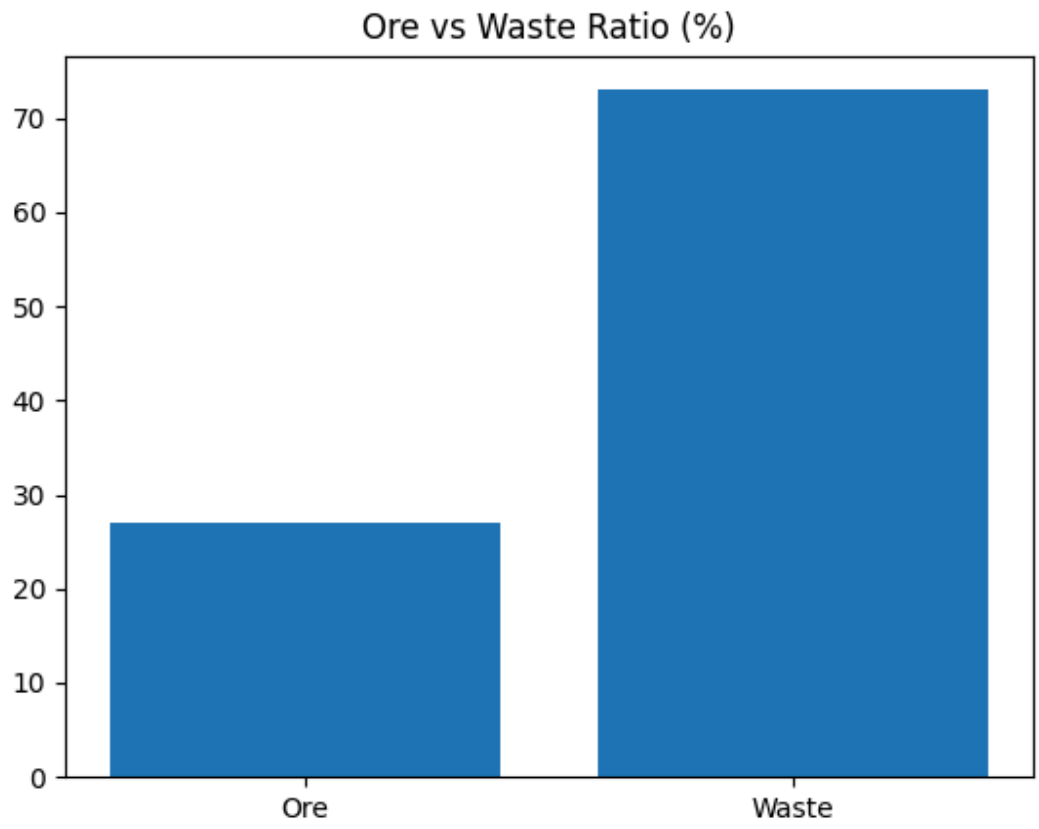
2. Ore vs Waste Ratio



This graph represents the proportion of ore extracted compared to waste material removed. Large open-pit gold mines require significant waste removal.
👉 **Significance:** Important for **stripping ratio analysis, cost estimation, and mine design optimization**.

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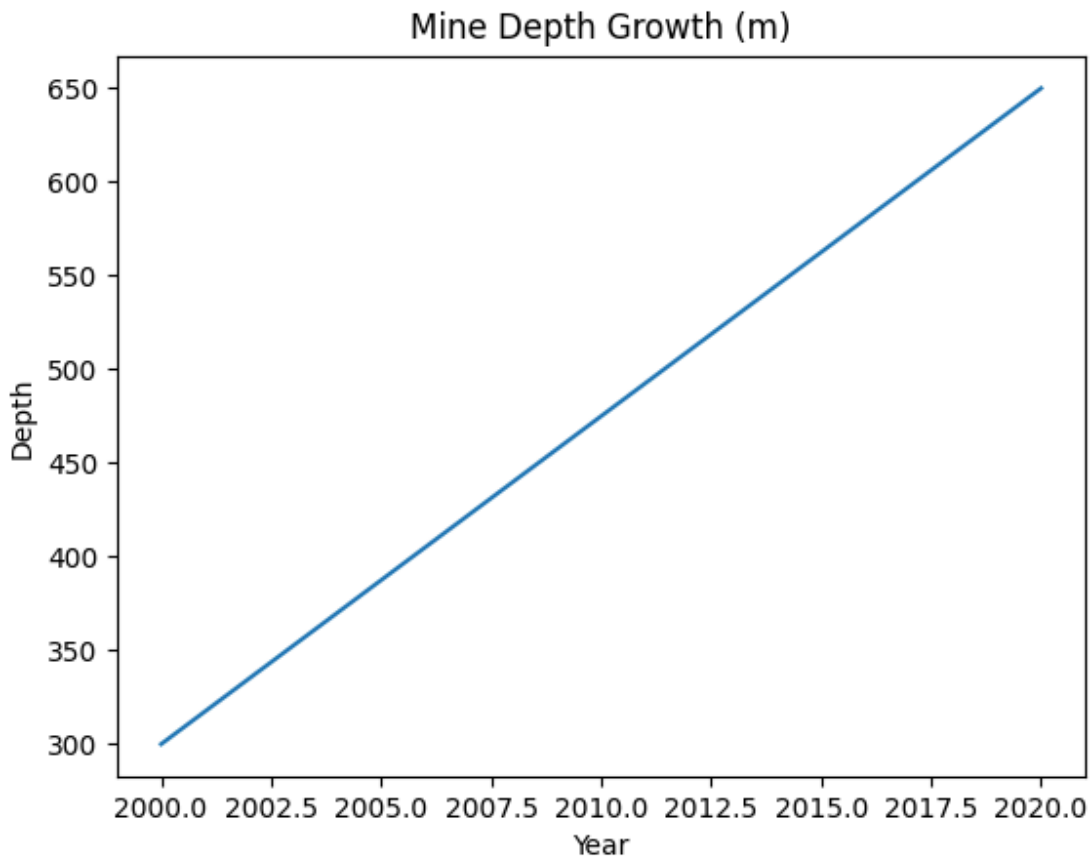
3. Mine Depth Growth (m)



This chart illustrates how the mine depth increases over time as extraction progresses. Greater depth increases operational challenges.
👉 **Significance:** Crucial for **slope stability, geotechnical design, and long-term mine planning**.

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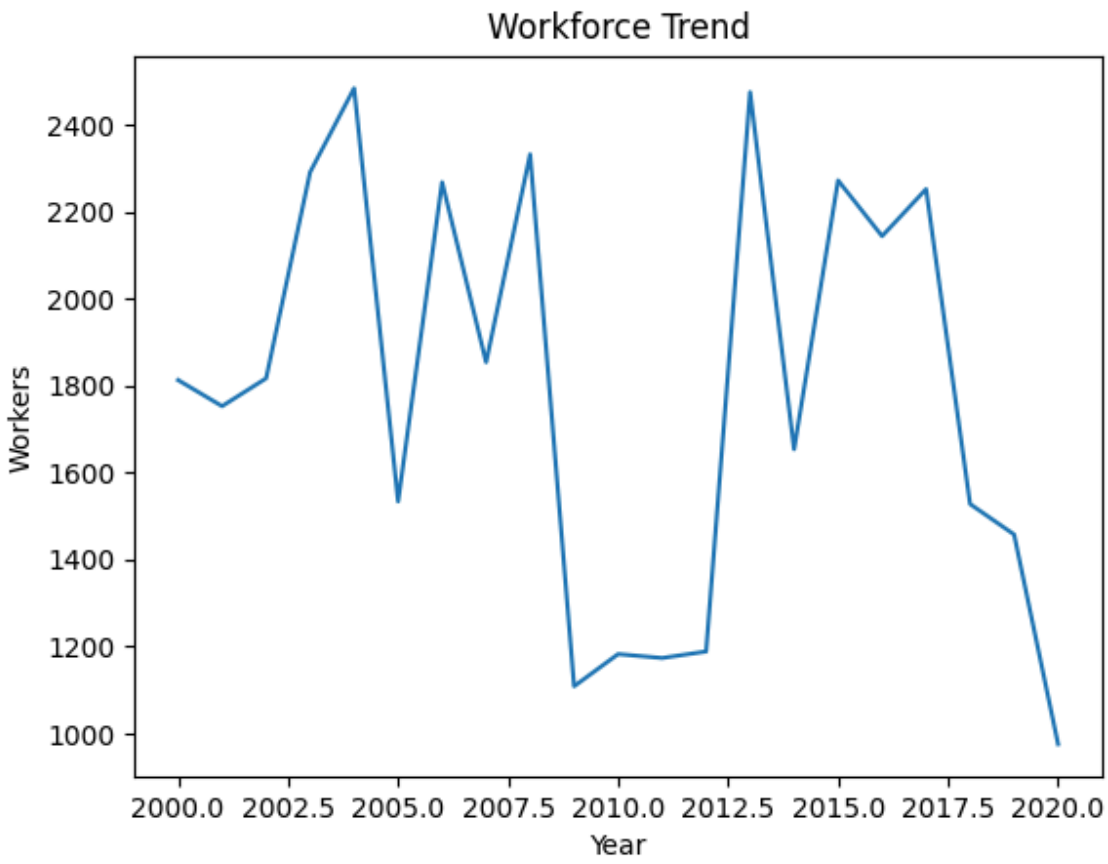
4. Workforce Trend



This graph shows the number of workers employed over time. Changes may reflect automation and operational scaling.
👉 **Significance:** Helps in **labour planning, productivity analysis, and safety management**.

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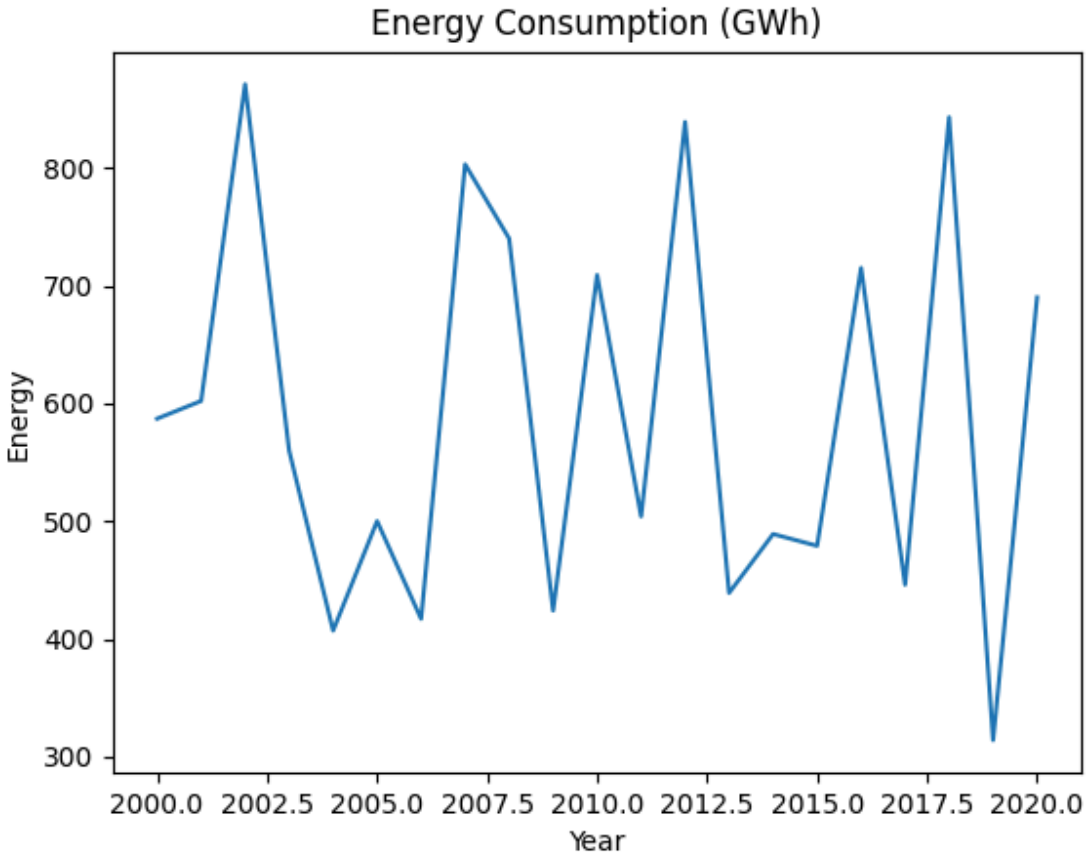
5. Energy Consumption (GWh)



This chart displays energy usage for mining and processing operations, including crushing and leaching.
👉 **Significance:** Useful for **energy efficiency, cost control, and sustainable mining practices**.

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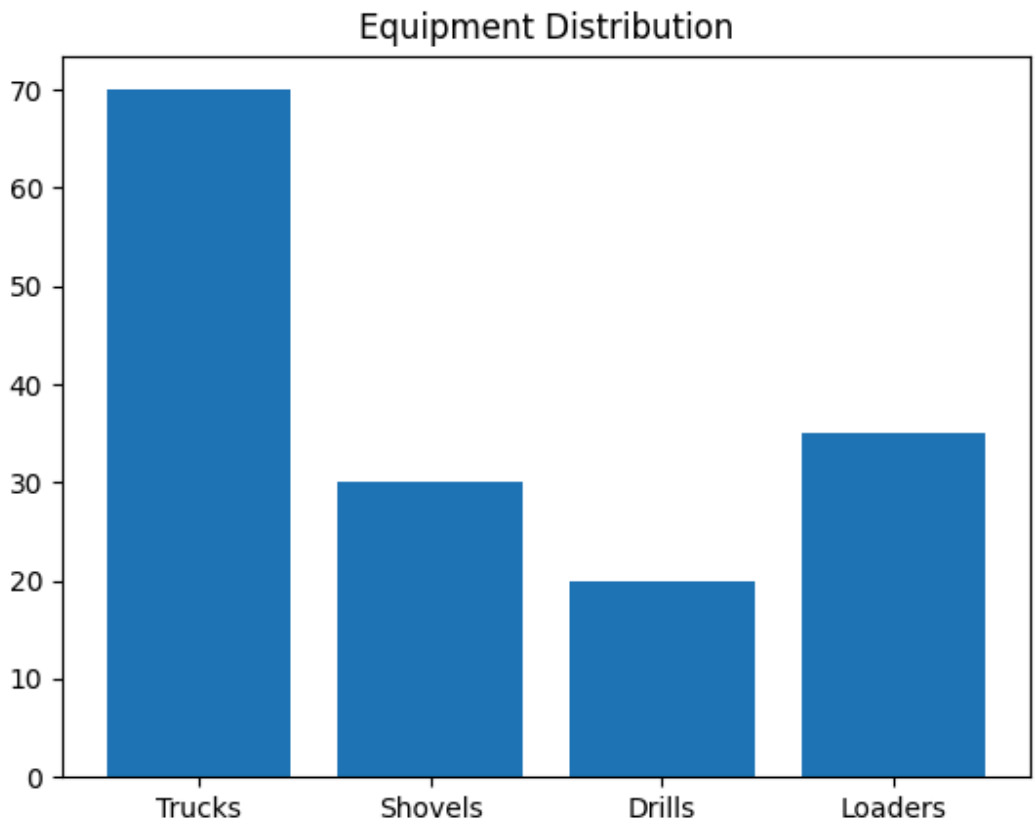
6. Equipment Distribution



This chart shows the types and quantities of heavy machinery used, such as trucks and shovels.
👉 **Significance:** Important for **fleet management, equipment selection, and maintenance planning**.

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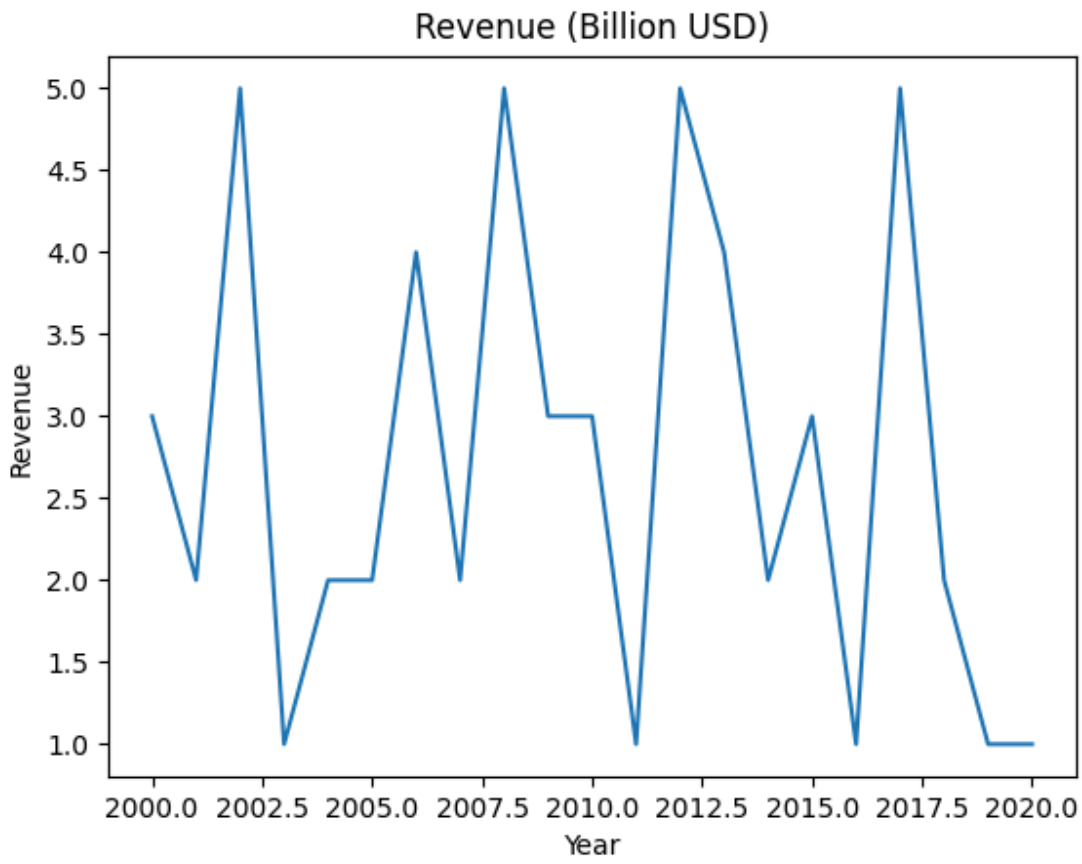
7. Revenue (Billion USD)



This graph shows the annual revenue generated from gold production. It depends on production levels and global gold prices.
👉 **Significance:** Essential for **economic evaluation, financial planning, and investment decisions**.

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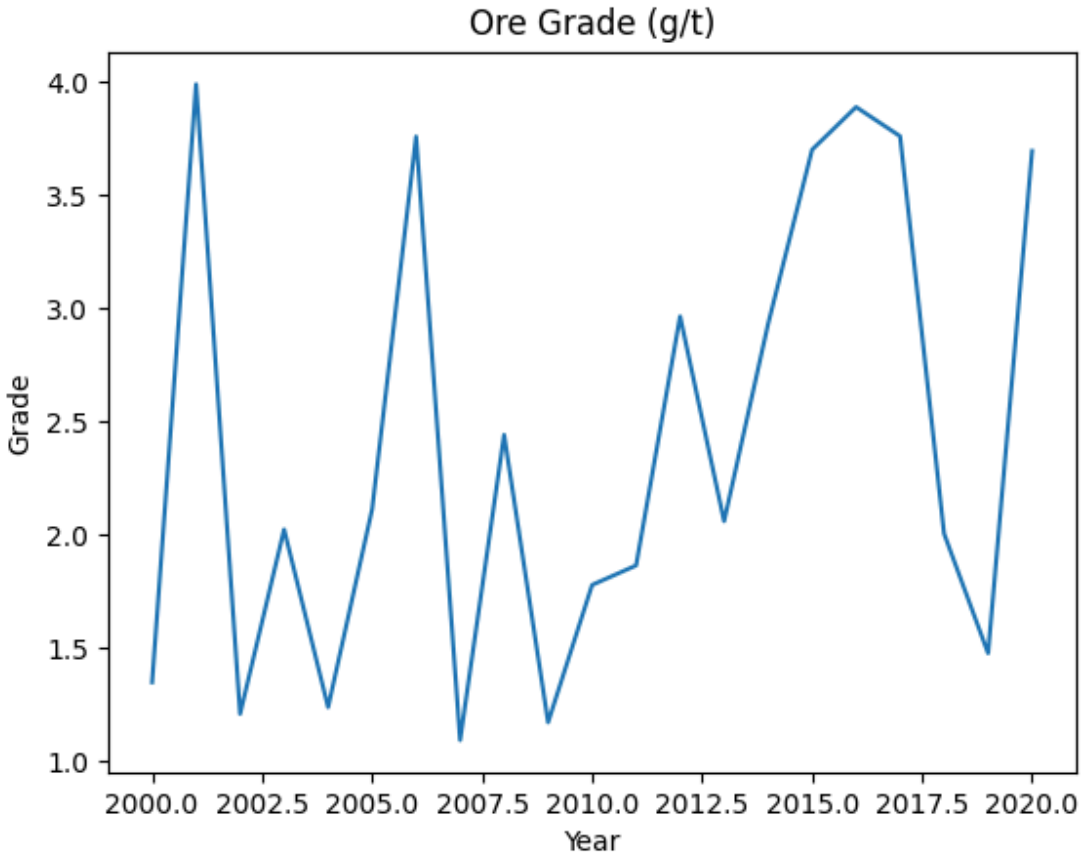
8. Ore Grade (g/t)



This chart represents the gold concentration in the ore. Higher grades result in more efficient gold extraction.
👉 **Significance:** Directly affects **processing efficiency, recovery rates, and profitability**.

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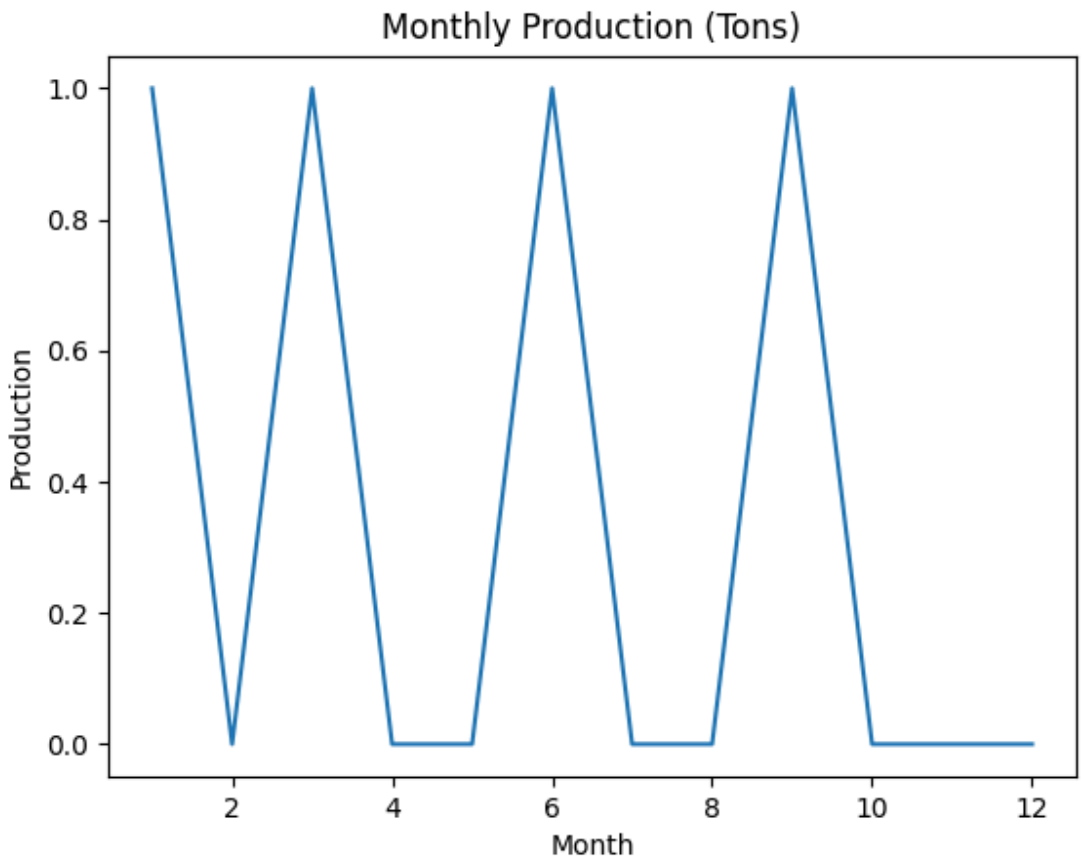
9. Monthly Production (Tons)



This graph shows variation in production throughout the year. Changes may be due to operational or maintenance factors.
👉 **Significance:** Helps in **short-term planning, scheduling, and performance monitoring**.

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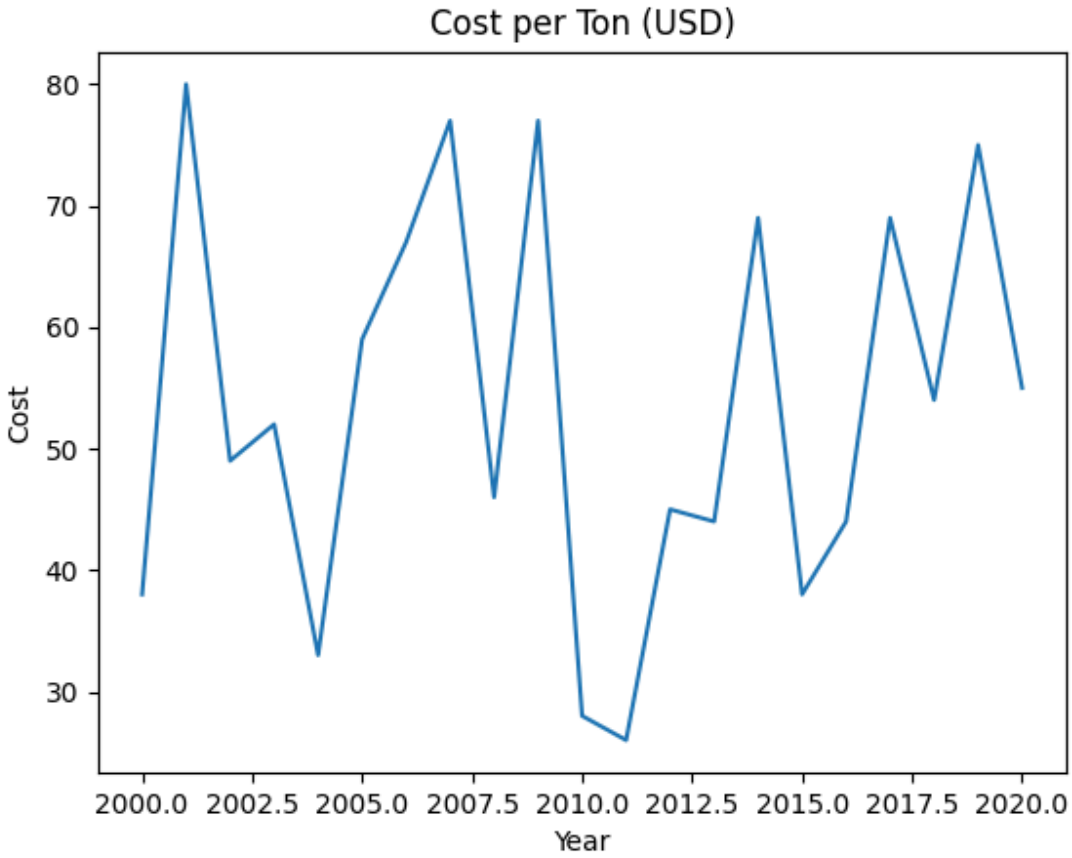
10. Cost per Ton (USD)



This chart shows the cost of extracting and processing one ton of material. Costs are influenced by depth, energy use, and equipment efficiency.
👉 **Significance:** Critical for **cost optimization, budgeting, and improving operational efficiency**.